Solution of Question 1.

(a) Apply the Euclidean algorithm to the polynomials x^4+x+1 and x^2+1. One has

$$\begin{aligned} x^4+x+1 &= \left(x^2+1\right)\cdot\left(x^2+1\right)+x \\ x^2+1 &= x\cdot x+1. \end{aligned}$$

As a consequence, [3/5]

$$\begin{aligned} 1 &= \left(x^2+1\right)+x\cdot x \\ &= \left(x^2+1\right)+x\cdot\left(\left(x^4+x+1\right)+\left(x^2+1\right)\left(x^2+1\right)\right) \\ &= x\left(x^4+x+1\right)+\left(x^3+x+1\right)\left(x^2+1\right). \end{aligned}$$

Therefore,

$$\begin{aligned} a(x) &= 1, \\ b(x) &= x^3+x+1. \end{aligned}$$

[2/5]

(b)

i

$$\boldsymbol{H}=\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

[2]

ii

$$\boldsymbol{G}=\begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

[2]

iii The syndrome vector is given by

$$\boldsymbol{s}_1=\boldsymbol{y}_1\boldsymbol{H}^T=[0\,1\,1],$$

hence

$$\hat{\boldsymbol{c}}_1=[0\,0\,1\,1\,1\,0\,0],$$

and

$$\hat{\boldsymbol{m}}_1 = [1\ 1\ 0\ 0]\,.$$

[3]

iv The syndrome vector is given by

$$[0\ 1\ 1\ 0\ 0\ 0\ 1]\,\boldsymbol{H}^T = [1\ 1\ 0]\,.$$

Let c_0 and c_4 be the missing symbols in $\boldsymbol{c}$. Then one has

$$[c_0\ c_4]\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = [1\ 1\ 0]\,.$$

It is clear that $[c_0\ c_4] = [0\ 1]$. Hence $\boldsymbol{c}_2 = [0\ 1\ 1\ 0\ 1\ 0\ 1]$ and $\boldsymbol{m}_2 = [0\ 1\ 0\ 1]$. [3]

(c)

i Here note that $0 - 1 = 2 \in \mathbb{F}_3$. One has

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

[2]

ii From the fact that if $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}]$ then $\boldsymbol{G} = \left[-\boldsymbol{A}^T\ \boldsymbol{I}\right]$. One has

$$\boldsymbol{G} = \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

[2]

(d)

i Note that $x^2 + 1 = (x+1)(x+1)$ in $\mathbb{F}_2[x]$. Hence $x^2 + 1 \in \mathbb{F}_2[x]$ is not irreducible. [2]

ii Note that $x^2 + 1 = (x+1)(x+2) + 2$ in $\mathbb{F}_3[x]$. There is no polynomial with degree one that divides $x^2 + 1$ in $\mathbb{F}_3[x]$. $x^2 + 1 \in \mathbb{F}_3[x]$ is irreducible. [2]

(e)

$$(x+1)^p = \sum_{i=0}^{p} \binom{p}{i} x^i.$$

It is clear that $\binom{p}{0} = \binom{p}{p} = 1$. It suffices to prove that $p | \binom{p}{i}$ for all $1 \leq i \leq p-1$. Note that

$$\binom{p}{i} = \frac{p\,(p-1)\cdots(p-i+1)}{i!} \in \mathbb{Z}.$$

That is

$$(i!)\,|p \cdot ((p-1)\cdots(p-i+1))\,.$$

Because $\gcd(i!, p) = 1$, by Euclid's Lemma one has $(i!) \mid (p-1)\cdots(p-i+1)$. Hence $p | \binom{p}{i}$ for all $1 \leq i \leq p-1$. [2]

Solutions of Question 2.

(a) The Singleton bound for linear codes $\mathcal{C}[n, k, d]$ states that

$$d \leq n - k + 1.$$

To prove it, consider the corresponding parity check matrix $\boldsymbol{H} \in \mathbb{F}_q^{(n-k)\times n}$. [1/3]
The code distance d is defined as the minimum number of linearly dependent columns of $\boldsymbol{H}$. On the other hand, for the parity check matrix $\boldsymbol{H} \in \mathbb{F}_q^{(n-k)\times n}$, every $n - k + 1$ columns have to be linearly dependent. Therefore,

$$d \leq n - k + 1.$$

[2/3]

(b)

i Let $\boldsymbol{c}_1 = \text{eval}(f_1)$ and $\boldsymbol{c}_2 = \text{eval}(f_2)$ where deg $f_1 \leq k-1$ and deg $f_2 \leq k-1$. Then $\alpha\boldsymbol{c}_1 + \beta\boldsymbol{c}_2 = \text{eval}(g)$ with $g = \alpha f_1 + \beta f_2$. Since deg $g \leq k-1$, eval $(g) \in \mathcal{C}$. [3]

ii A polynomial of degree $\leq k-1$ can have at most $k-1$ zeros. Hence, $\forall \boldsymbol{c} \in \mathcal{C}$ s.t. $\boldsymbol{c} \neq 0$, $\boldsymbol{c} = \text{eval}(f)$ has weight at least $n - k + 1$. [3]

iii Based on the definition, the generator matrix of a Reed-Solomon code is given by

$$\boldsymbol{G} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \alpha & \cdots & \alpha^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{k-1} & \cdots & \alpha^{(k-1)(n-1)} \end{bmatrix}.$$

[3]

iv It is clear that $\boldsymbol{s} = \boldsymbol{y}\boldsymbol{H}^T = (\boldsymbol{c} + \boldsymbol{e})\boldsymbol{H}^T = \boldsymbol{e}\boldsymbol{H}^T$. Hence

$$s_j = \sum_{i=1}^{n} e_i h_{j,i} = \sum_{i \in \mathcal{I}} e_i \alpha^{(i-1)j}.$$

[2]

v By definition,

$$\begin{aligned} S(z) &= \sum_{j=0}^{n-k-1} s_{j+1} z^j \\ &= \sum_{j=0}^{n-k-1} \left(\sum_{i \in \mathcal{I}} e_i \alpha^{(i-1)(j+1)} \right) z^j \\ &= \sum_{i \in \mathcal{I}} e_i \alpha^{i-1} \left(\sum_{j=0}^{n-k-1} \left(\alpha^{i-1} z \right)^j \right) \\ &\equiv \sum_{i \in \mathcal{I}} e_i \alpha^{i-1} \left(\sum_{j=0}^{\infty} \left(\alpha^{i-1} z \right)^j \right) \mod z^{n-k} \\ &= \sum_{i \in \mathcal{I}} \frac{e_i \alpha^{i-1}}{1 - \alpha^{i-1} z}. \end{aligned}$$

[5]

vi The key equation used for Reed-Solomon decoding is given by

$$E(z) = L(z) \cdot S(z) \mod z^{n-k}.$$

[2]

vii Apply Euclidean algorithm to the polynomials z^{n-k} and $S(z)$ until the the degree of the remainder polynomial is at most $\left\lfloor \frac{n-k}{2} \right\rfloor - 1$. [1/4]
According to Bézout's identity, one has

$$a(z) = b(z) S(z) + c(z) z^{n-k},$$

for some polynomials $a(z)$, $b(z)$, and $c(z)$. [2/4]
Denote the constant term of $b(z)$ by d. Then $L(z) = d^{-1} b(z)$ and $E(z) = d^{-1} a(z)$. [1/4]

Solutions of Question 3.

(a)

i As all elements in $\beta\mathbb{F}_q^*$ has the form $\beta\beta_i \neq 0$, it is clear that $\beta\mathbb{F}_q^* \subseteq \mathbb{F}_q^*$. For any given $\beta_i \neq \beta_j$, $\beta(\beta_i - \beta_j) \neq 0$ or equivalently $\beta\beta_i \neq \beta\beta_j$. Therefore, $\left|\beta\mathbb{F}_q^*\right| = q-1$. The only possibility is therefore $\beta\mathbb{F}_q^* = \mathbb{F}_q^*$. [2]

ii Since $\beta\mathbb{F}_q^* = \mathbb{F}_q^*$, it holds that

$$\prod_{\gamma\in\beta\mathbb{F}_q^*} \gamma = \prod_{\gamma\in\mathbb{F}_q^*} \gamma.$$

That is

$$\beta^{q-1} \prod_{\gamma\in\mathbb{F}_q^*} \gamma = \prod_{\gamma\in\mathbb{F}_q^*} \gamma.$$

Note that $\prod_{\gamma\in\mathbb{F}_q^*} \gamma$ is a nonzero element and its multiplicative inverse exists. Hence, $\beta^{q-1} = 1$. [2]

(b) Let $t = \text{ord}(\beta)$. Suppose that $t \nmid (q-1)$. Then there exists $0 < r < t$ such that $(q-1) = \ell \cdot t + r$ for an integer ℓ. By Fermat's Little Theorem, $1 = \beta^{q-1} = \beta^{\ell\cdot t+r} = \beta^r$. This contradicts the definition that $t = \text{ord}(\beta)$ is the smallest integer such that $\beta^t = 1$. Hence $t|(q-1)$. [3]

(c)

i The complete table is given by

x	1	2	5	10	31	62	155
$2^x \bmod 311$	2	4	32	91	**36**	**52**	**1**
$17^x \bmod 311$	17	289	142	260	**305**	**36**	**310**

The detailed computations are given as follows.

$\text{mod}(2^{31}, 311) = \text{mod}\left((2^{10})^3 \cdot 2, 311\right) = \text{mod}(91^3 \cdot 2, 311) = 36.$

$\text{mod}(2^{62}, 311) = \text{mod}\left((2^{31})^2, 311\right) = \text{mod}(36^2, 311) = 52.$

$\text{mod}(2^{155}, 311) = \text{mod}\left((2^{31})^5, 311\right) = \text{mod}(36^5, 311) = 1.$

$\text{mod}(17^{31}, 311) = \text{mod}\left((17^{10})^3 \cdot 17, 311\right) = \text{mod}(260^3 \cdot 17, 311) = 305.$

$\text{mod}(17^{62}, 311) = \text{mod}\left((17^{31})^2, 311\right) = \text{mod}(305^2, 311) = 36.$

$\text{mod}(17^{62}, 311) = \text{mod}\left((17^{31})^5, 311\right) = \text{mod}(305^5, 311) = 310.$ [6]

ii The number 17 is a primitive element in $\mathbb{F}_{311}$.

The number 2 is clearly not a primitive element in $\mathbb{F}_{311}$ as $2^{155} = 1$. [2/4]

The number 17 is a primitive element because $\text{ord}(17) = 310$. To check that, note that $310 = 2 \times 5 \times 31$. The only possible values for $\text{ord}(17)$ are $1, 2, 5, 10, 31, 62, 155, 310$. From the third line of the above table, it is clear that $\text{ord}(17) = 310$. [2/4]

iii From the table, it is clear that

$$10 = \log_{17} 260 \bmod 311,$$

and

$$62 = \log_{17} 36 \bmod 311.$$

[2]

(d)

i It is clear that $t = 310 \times 312 = 96720$. Apply Euclidean algorithm. One has

$$\begin{aligned} 96720 &= 23 \times 4205 + 5 \\ 23 &= 5 \times 4 + 3 \\ 5 &= 3 + 2 \\ 3 &= 2 + 1. \end{aligned}$$

Based on Bézout's identity, one has

$$\begin{aligned} 1 &= 3 - 2 = 3 - (5 - 3) = 3 \cdot 2 - 5 \\ &= (23 - 5 \cdot 4) \cdot 2 - 5 = 23 \cdot 2 - 5 \cdot 9 \\ &= 23 \cdot 2 - (96720 - 23 \cdot 4205) \cdot 9 \\ &= -96720 \cdot 9 + 23 \cdot 37847. \end{aligned}$$

Hence

$$d = 37847.$$

[3]

ii

$$c = 11^{23} \bmod 97343.$$

Note that $23 = 16 + 4 + 2 + 1$. Furthermore,

x	1	2	4	8	16
$11^x \bmod 97343$	11	121	14641	9595	74890

$11^{20} = 11^{16} \cdot 11^4 = 90281 \bmod 97343$, and $11^{23} = 90281 \cdot 121 \cdot 11 = 42749 \bmod 97343$. Therefore,

$$c = 42749.$$

[3]